Arrays:

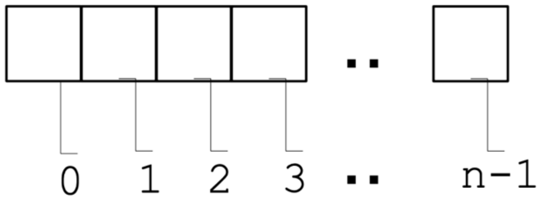
An array is an indexed list of values that all have the **same type**.

You can make an array of any type int, double, String, etc.

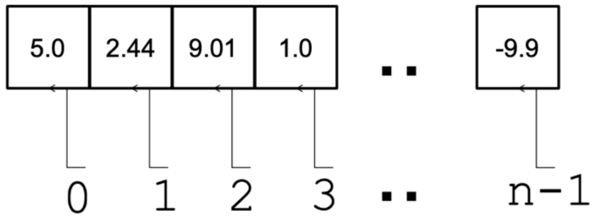
In array, the memory allocated is contiguous.

1. One-dimensional Array:

It is an array that has one index. **The index starts at zero and ends at length-1.**



Following an example of array of type double:



How to declare, initialize and access array?

An array is **defined** as following:

data_type array_name[];

OR

data_type[] array_name;

```
int[] values; // declare array of type int named “values”
int values[]; // same as above
```

To create an array of a given size, here size specifies the number of elements in the array, use the operator **new**:

```
double[] c = new double[5];// declare an double array named “c” of size
5
double c[] = new double[5];// same as above
```

Curly braces { } can be used to **initialize** an array. It can **ONLY** be used when you declare the variable.

```
int[] values = { 12, 24, -23, 47 };// size of array deduced from the
number of items
int values[] = { 12, 24, -23, 47 };// same as above
```

Each array has a **length** variable built-in that contains the length of the array.

```
int[] values1 = new int[12]; // Declare and allocate a 12 elements int array
int num = values1.length; // num is 12

int[] values2 = { 1, 2, 3, 4, 5 }
int size = values2.length; // size is 5
```

To access the elements of an array, use the [ ] operator. This operator contains the index value.

```
int[] values = { 12, 24, -23, 47 };
values[3] = 18; // {12, 24, -23, 18}
int x = values[1] + 3; // x = 27
```

Note: If the index is either outside the range of [0, length-1] or negative, Java Runtime will signal an exception called **ArrayIndexOutOfBoundException**.

Q> Is there an error in the following codes?

1. `int[] values = {1, 2.5, 3, 3.5, 4};`
2. `char[] array = {'a', '*', '4', '/'};`
   `System.out.println(array[4]);`

You can process all the elements of an array via a **loop.**

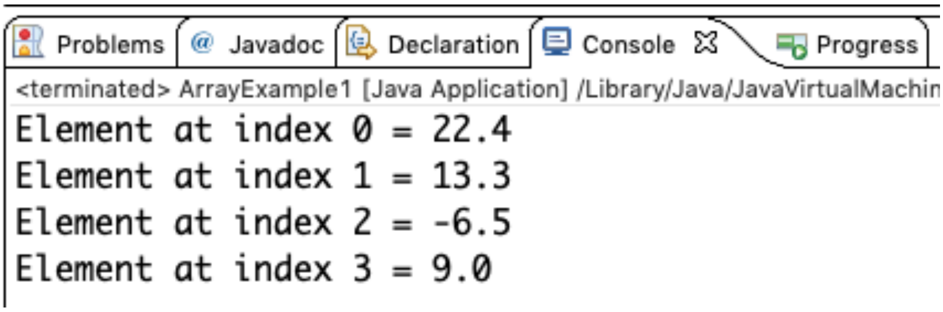
```
int[] values = { 12, 24, -23, 47 };
for (int i=0; i < values.length; i++)
{
System.out.println(values[i]);
}
```

```
12
24
-23
47
```

```
int[] values = { 12, 24, -23, 47 };
int i = 0;
while (i < values.length)
{
System.out.println(values[i]);
i++;
}
```

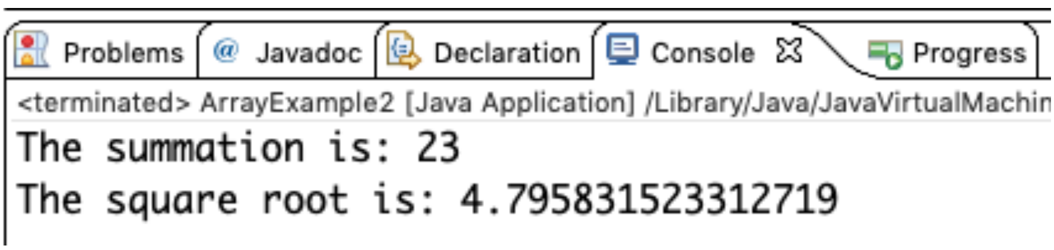
```
12
24
-23
47
```

```java
/* Java program to illustrate creating an array of type double and
   print each element with its index */
public class ArrayExample1 {
     public static void main(String[] args) {
     double[] array= {22.4, 13.3, -6.5, 9};
     for (int c = 0; c < array.length; c++)
     System.out.println("Element at index "+ c + " = " + array[c]);
     }
}
```

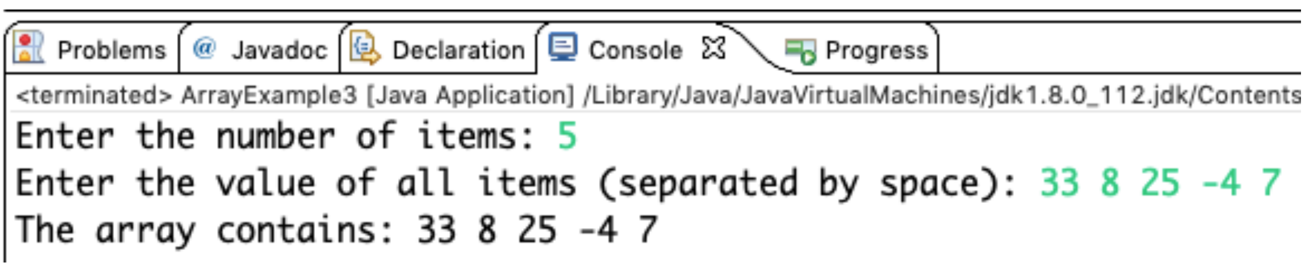


Example: Write a Java program to find the summation of the elements in an array then print out the square root of that summation.

```java
public class ArrayExample2 {
     public static void main(String[] args) {
     int numbers[] = {8, 2, 6, 4, 3};
     int sum = 0;
     for (int i=0; i < numbers.length; i++)
          sum = sum + numbers[i];
     System.out.println("The summation is: " + sum);
     System.out.println("The square root is: " + Math.sqrt(sum));
     }
 }
```



```
// Java Example Read and Print Array
import java.util.Scanner;
public class ArrayExample3 {
   public static void main(String[] args) {

     // Declare variables
       int size;
       int[] items;
       Scanner in = new Scanner(System.in);

     // Enter the number of items i.e. the size of the array
       System.out.print("Enter the number of items: ");
       size = in.nextInt();

       items = new int[size]; // Allocate the array

       // Read the items of the array
       System.out.print("Enter the value of all items (separated by space): ");
       for (int i = 0; i < items.length; i++)
          items[i] = in.nextInt();

       in.close();

     // Print array contents
       System.out.print("The array contains: ");
       for (int i = 0; i < items.length; i++)
       System.out.print(items[i] +" ");
   }
}
```



Example 1: Write a Java program to store and display the maximum daily temperatures in a week.

```java
import java.util.Scanner;
public class TemperatureReadings {
	public static void main(String[] args) {
	Scanner keyboard = new Scanner(System.in);
	double[] temperature; // Create the array
	temperature = new double[7]; // Allocate a 7 elements
	// Enter temperatures
	for (int i = 0; i < temperature.length; i++)
	{
	System.out.println("enter max temperature for day " + (i+1));
	temperature[i] = keyboard.nextDouble();
	}
	keyboard.close();
	// Display Temperatures
	System.out.println(); // blank line
	System.out.println("***TEMPERATURES ENTERED***");
	System.out.println(); // blank line
	for (int i = 0; i < temperature.length; i++)
		System.out.println("day "+(i+1)+" "+ temperature[i]);
	}
}
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> TemperatureReadings [Java Application] /Library/Java/JavaVirtuall
enter max temperature for day 1
23
enter max temperature for day 2
25
enter max temperature for day 3
22
enter max temperature for day 4
26
enter max temperature for day 5
27
enter max temperature for day 6
25
enter max temperature for day 7
30

***TEMPERATURES ENTERED***

day 1 23.0
day 2 25.0
day 3 22.0
day 4 26.0
day 5 27.0
day 6 25.0
day 7 30.0
```

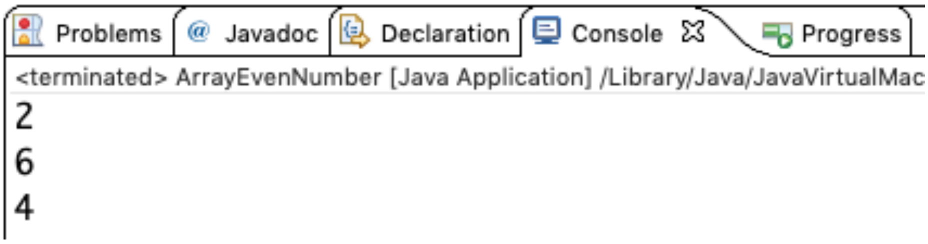
Example 2: Write a Java program to store a week temperature and print the highest temperature.

```java
import java.util.Scanner;
public class HighestTemperature {
	public static void main(String[] args) {
	Scanner keyboard = new Scanner(System.in);
	double[] temperature; // Create the array
	temperature = new double[7]; // Allocate a 7 items in the array
	// Enter Temperatures
	for (int i = 0; i < temperature.length; i++)
	{
	System.out.println("enter max temperature for day " + (i+1));
	temperature[i] = keyboard.nextDouble();
	}
	keyboard.close();

	double highest=temperature[0];// Assume the first item is the highest
	//This loop runs from the 2nd item to the last item in the array
	for (int i = 1; i < temperature.length; i++)
		if (temperature[i] > highest)
			highest=temperature[i]; // Reset result to new highest
	System.out.println("The highest temperature of the week " + highest);
	}
}
```

```
Problems  Javadoc  Declaration  Console  Progress
<terminated> HighestTemperature [Java Application] /Library/Java/JavaVirtualM
enter max temperature for day 1
25
enter max temperature for day 2
27
enter max temperature for day 3
26
enter max temperature for day 4
30
enter max temperature for day 5
29
enter max temperature for day 6
28
enter max temperature for day 7
24
The highest temperature of the week 30.0
```

Example 3: Write a Java program to print even numbers in an array.

```
public class ArrayEvenNumber {
     public static void main(String[] args) {
     int numbers[] = {5, 2, 6, 3, 4};
     for (int i=0; i < numbers.length; i++)
          if (numbers[i] %2 == 0)
               System.out.println(numbers[i]);
     }
}
```



Example 4: Write a Java program to print the average of numbers in an array.

```
public class ArrayAverage {
     public static void main(String[] args) {
     double[] array= {22.4, 13.3, 9.5, 6, 10.3};
     double sum=0;
     for (int c = 0; c < array.length; c++)
          sum=sum+array[c];
     System.out.println("The average = " + sum/array.length);
     }
}
```

Example 5: Write a Java program to search an element in an array.

```
import java.util.Scanner;
public class ArraySearch {
     public static void main(String[] args) {
     int numbers[] = {5, 2, 7, 4, 10};
     int i,num,flag=0;
     Scanner s = new Scanner(System.in);
     System.out.print("Enter the element you search for : ");
     num = s.nextInt();
     s.close();

     for (i=0; i < numbers.length; i++)
          if (num==numbers[i])
          {
               flag=1;
               break;
          }
```

```java
      if (flag==1)
            System.out.println("The Element Found at position: "+ i);
      else
            System.out.println("The Element Not Found");
      }
}
```

```
Problems  @ Javadoc  Declaration  Console ⌧  Progress
<terminated> ArraySearch [Java Application] /Library/Java/JavaVirtualMachines/
Enter the element you search for : 7
The Element Found at position: 2
```

```
Problems  @ Javadoc  Declaration  Console ⌧  Progress
<terminated> ArraySearch [Java Application] /Library/Java/JavaVirtualMachines/
Enter the element you search for : 3
The Element Not Found
```

Example 6: Write a Java program to sort the items of an array in ascending order.

```java
public class ArrayAscendingOrder {
      public static void main(String[] args) {
            double[] array= {22.4, 13.3, 9.5, 6, 10.3};
            double temp;

          for (int i = 0; i < array.length; i++)
          {
              for (int j = i + 1; j < array.length; j++)
              {
                  if (array[i] > array[j])
                  {
                      temp = array[i];
                      array[i] = array[j];
                      array[j] = temp;
                  }
              }
          }
          System.out.print("Ascending Order: ");
          for (int i = 0; i < array.length; i++)
            System.out.print(array[i] + " ");
    }
}
```

```
Problems  @ Javadoc  Declaration  Console ⌧  Progress
<terminated> ArrayAscendingOrder [Java Application] /Library/Java/JavaVirtualM
Ascending Order: 6.0 9.5 10.3 13.3 22.4
```

Homework:

Q> Write a Java program to count the number of occurrences of an element in an array.

Q> Write a Java program to print the prime numbers in an array.

Q> Write a Java program to sort the items of an array in descending order.

Q> Write a Java program to sort names of an array in an alphabetical order.